1. Sum of fourth powers

AIM: To compute the sum of fourth powers of numbers from 1 to 100 whose digit sum is divisible by 5.

```
result = sum(n**4 for n in range(1, 101) if sum(int(d) for d in
str(n)) % 5 == 0)
result

387559626
```

RESULT: Thus, the given Python program was successfully executed.

2. Count numbers with repeated digits

AIM: To count the numbers between 1 and 500 that contain at least one repeated digit.

```
count = sum(1 for n in range(1, 501) if len(set(str(n))) <
len(str(n)))
count

122
```

RESULT: Thus, the given Python program was successfully executed.

3. Count words having length greater than 6

AIM: To count the total number of words having length greater than 6 in a list of sentences.

```
sen = [
    "He kicked the football across the grass",
    "She watches a football match every weekend",
    "The Olympics is the ultimate athletic tournament"
]

count = sum(1 for s in sen for word in s.split() if len(word) > 6)
count

8
```

RESULT: Thus, the given Python program was successfully executed.

4. Generate Pythagorean triples

AIM: To generate all ordered triples satisfying $x^2 + y^2 = z^2$ for values from 1 to 5.

```
triples = ((x, y, z)
    for x in range(1, 6)
    for y in range(1, 6)
    for z in range(1, 6)
    if x*x + y*y == z*z
)

print(list(triples))
```

```
[(3, 4, 5), (4, 3, 5)]
```

RESULT: Thus, the given Python program was successfully executed.

5. Sum of numbers divisible by 3 or 7 but not both

AIM: To compute the sum of numbers below 200 that are divisible by 3 or 7 but not both.

```
result = sum(n for n in range(200) if (n % 3 == 0) ^ (n % 7 == 0))
result
```

```
7585
```

RESULT: Thus, the given Python program was successfully executed.

6. Sum of elements above the main diagonal

AIM: To compute the sum of elements above the main diagonal of a matrix.

```
matrix = [[2,4,6],
          [1,3,5],
          [7,8,9]]

total = sum(
    matrix[i][j]
    for i in range(len(matrix))
    for j in range(len(matrix))
    if j > i
)
```

```
total

15
```

RESULT: Thus, the given Python program was successfully executed.

7. Count words containing exactly two vowels

AIM: To count the number of words containing exactly two vowels.

```
words = ["quantum", "gravity", "friction", "atom", "velocity",
"energy"]
vowels = "aeiou"

count = sum(1 for w in words if sum(1 for ch in w if ch in vowels) ==
2)

count

3
```

RESULT: Thus, the given Python program was successfully executed.

8. Flatten dictionary values and find sum

AIM: To flatten the nested dictionary values and compute their total sum.

```
data = {
    "A": [1,2,3],
    "B": [4,5],
    "C": [6,7,8]
}

total = sum(x for values in data.values() for x in values)

total

36
```

RESULT: Thus, the given Python program was successfully executed.

9. Sum of factorials from 1 to 8

AIM: To compute the sum of factorials of numbers from 1 to 8 using a generator expression.

```
import math

result = sum(math.factorial(n) for n in range(1, 9))
result

46233
```

RESULT: Thus, the given Python program was successfully executed.

10. Count numbers whose square ends with 6

AIM: To count the numbers from 1 to 300 whose square ends with the digit 6.

```
count = sum(1 for n in range(1, 301) if (n*n) % 10 == 6)
count

60
```

RESULT: Thus, the given Python program was successfully executed.